

Name:

Stud.ID:

Vision and Cognitive Systems

A.Y. 2021/2022

Midterm Exam - December 6, 2021

Exam Instructions

- Write your *name* and *Stud.ID (matricola)* on **every** page.
- You have just one page (one side) to answer a question.
- You will have **exactly** 90 min to complete your exam.

Question 1 (3 points)

Describe in details a bag-of-visual-words pipeline. What are its main stages and what are their roles?

Question 2 (3 points)

What's the purpose of image filtering? You should give a clear definition of filtering and also define the mathematical tools used in order to define any filter. Finally, give an example of a smoothing filter (and provide also its definition).

Question 3 (2 points)

What's a perceptron? Describe in details its main components and how a multi-layer perceptron can add more representation power w.r.t. a simple perceptron.

Question 4 (2 points)

What is this filter $\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$? Please “decompose” it in its two 1D components and highlight their respective roles.